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MILITARY SPECIFICATION
TEST STATION, LOW FREQUENCY
OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Low Frequency Test Station (LFTS) OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(V) (see 6.9).

1.2 Classification. The LFTS consists of a 3-bay station, an antenna mounting stand, and accessories as follows (see 6.4):

a. Bay No. 1: Console, Test, Low Frequency, Receiver-hpllfier OJ-253( )/AWM-23(V).

	Nomenclature	Reference Paragraph
(1)	Panel, Power Distribution SB-3696( )/AWM-23(V)	3.6.1.1
(2)	Counter/Timer	3.6.1.2
(3)	Digital Multimeter	3.6.1.3
(4)	Monitor, Coordinate Data ID-1861( )/AWM-23(V)	3.6.1.4
(5)	Frequency Response Analyzer	3.6.1.5
(6)	Panel, Coolant Supply MX-9307( )/AWM-23(V)	3.6.1.6
(7)	Mounting MT-4502( )/AWM-23(V)	3,6.1.7
(8)	Power Supply PP-6850( )/AWM-23(V)	3.6.1.8
(9)	Power Supply, DC	3.6.1.9
(10)	Distribution Box J-3110( )/AWM-23(V)	3.6.1.10

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4935

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b. Bay No. 2: Console, Test, Low Frequency, Antenna OJ-254( )/ AWM-23(V)* and OJ- 380( )/ AWM-23 (V)**

Nomenclature	Reference Paragraph
1) Oscilloscope	5.6.2.1
2) Panel, Test, Electrical C-9179( )/AWM-23(V)	3.6.2.2
(3) Switching Unit, Fault Isolation SA-1895( )/AWM-23(V)	3.6.2.3
(4) Switching Unit, Fault Isolation SA-1896( )/AWM-23(V)* or SA-2086( )/AWM-23(V)**	5.6.2.4

* Part of OJ-215( )/AWM-23(V) only.
** Part of OJ-379( )/AWM-23(V) only.

c. Bay No. 3: Console, Test, Low Frequency, Power Supply OJ-255( )/AWM-23(V)

Nomenclature	Reference Paragraph
(1) Spectrum Analyzer	3,6.3.1
(2) Switching Unit, Fault Isolation SA-1897( )/AWM-23(V)	3.6,3.2
(3) Panel, Coolant Supply MX-9308( )/AWM-23(V)	3.6.3.3
(4) Mounting MT-4503( )/AWM-23(V)	3.6.3.4
(5) Dummy Load, Electrical DA-631( )/AWM-23(V)	3.6.3.5
(6) Dummy Load, Electrical DA-632( )/AWM-23(V)	3.6.3.5
d. Stand, Mounting, Antenna MT-4445( )/AWM-23(V)	3.6.4
e. Accessories - LFTS	3.6.5
(1) Extenders - LFTS	3.6.5,1
(2) Fixture, Pressure Leakage - Test	3.6.5.2
(3) Fixture, Receiver/Refrigerator Cryogenic-Infrared*	3.6.5.3
(4) Fixture, Antenna Servo	3.6.5.4
(5) Fixture, Refrigerator Test*	3.6.5.5
(6) Equipment, Helium Filling*	3.6.5.6

* Used with OJ-215( )/AWM-23(V) only.

1.3 Associated equipment. The LFTS shall operate with the following associated equipment (see 6.4):

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a. Target Simulator, Infrared SM-630( )/AWM-23(V)*
b. Kit, Tool, Special TK-2Z2( )/AWN-25(V)
c. Adapter, Handling, Antenna MX-9046( )/AWM-23(V)
d. Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V)
e. Service Set, Hydraulic System MX-9043( )/AWM-23(V)
f. Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V)
g. Test Set, Antenna, Boresight TS-3568( )/AWM-23(V)

* Used with OJ-215( )/AWM-23(V) only.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWN-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the LFTS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System AN/AWM-23(V)

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

PUBLICATIONS

Hughes Aircraft Company

3282206	Acceptance Test Procedure, Low Frequency Test Station

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract - the contract shall have precedence over any other document.
b. This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).
c. Reference documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

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3.1 Materials, parts, and processes. Applicable.

3.2 Design and construction. Applicable except as follows.

3.2.1 Design features. Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Applicable.

3. 2.1.2,1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable,

3.2.1.7 Test set generated acoustic noise, Applicable.

3.2.1.8 Forced-liquid cooling. Applicable, except that the flow rate shall be 2.5 gallons per minute, maximum.

3.2.1.9 Space and weight limitations. The LFTS shall occupy less than 47.5 square feet of deck area. The weight of the LFTS shall be less than 3800 pounds including antenna stand. Maximum dimensions of the LFTS, excluding the antenna mounting stand and including the WRA mounting surfaces, shall be 73 inches by 48 inches by 75 inches with all drawers in the retracted position. Maximum dimensions for the antenna mounting stand shall be 50 inches by 50 inches by 53 inches, including an acoustic cover.

3.2. 1.9.1 Storage space. All items which are not permanently attached to the LFTS will be stored in separate storage facilities. A minimum of 12 cubic feet of storage space for the OJ-215( )/AWM-23(V) LFTS and 5 cubic feet of storage space for the OJ-379( )/AWM-23(V) LFTS will be required to store fixtures, test adapters, cables, manuals, and miscellaneous equipment.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. The total secondary electrical power shall be limited to 1600 watts maximum with any combination of power form loading within the load capability limits for each power form listed in Table I.

3.2. 1.11.1 Secondary power, semi- regulated Secondary semi-regulated power forms shall be provided by power supplies which are part of the LFTS. Power forms from 10 percent to full load and for normal values of utility line voltage variations shall be as listed in Table I.

3.2.1.11.2 Secondary power, regulated voltages Secondary regulated electrical power forms shall be provided by power supplies which are part of the

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TABLE I

SECONDARY POWER FORMS, SEMI-REGULATED

Item	Power Form (Vdc)	Load Capacity (Amperes)
1	$+6.5\ ^{+0.65}_{-1.4}$	5.0
2	$+9.7\ ^{+1.6}_{-1.8}$	45.0
3	$+14 \pm 1.4$	0.5
4	$+17\ ^{+2.8}_{-1.8}$	33.0
5	$+31\ ^{+5.1}_{-3.4}$	23.0
6	$+57\ ^{+5.7}_{-4.5}$	6.0
7	$+120\ ^{+12}_{-10}$	0.3
8	$-17\ ^{-2.8}_{+1.8}$	18.0
9	$-31\ ^{-5.1}_{+3.4}$	16.0
10	$-57\ ^{-5.7}_{+4.5}$	4.0
11	$-120\ ^{-12}_{+10}$	0.3

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LFTS. Regulation from no load to full load, and for normal values of facility line voltage variations shall not exceed 2 percent. Ripple voltage for direct current [de) power forms shall not exceed 150 millivolts (mV) peak-to-peak , (P-P). The power forms listed in Table II shall be provided.

3.2.1.12 Interface. The LFTS shall provide the Interfaces specified in the following,

3.2.1.12.1 Liquid cooler interface. The LFTS shall interface with Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V) through coolant lines which enter the LFTS at the top of bay no. 1.

3.2.1.12.2 Clean air interface. The LFTS shall interface with the clean air through ducts which enter the LFTS at the top rear of each test console.

3.2.1.12.3 Electrical Power interface. The LFTS shall interface with electrical power for the convenience outlets at one end of the convenience outlet strip mounted on the top of the front face of the LFTS. The LFTS shall interface with all other electrical power through power lines entering the top of bay number 1.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. Electrical power required for operation of the LFTS shall be as listed in Table III.

TABLE II

SECONDARY POWER FORMS, REGULATED

Item	Power Form	Load Capacity (amperes)*
1	+5 Vdc	20.0
2	+12 Vdc	0.8
3	+25 Vdc	3.0
4	+50 Vdc	3.5
5	-12 Vdc	0.3
6	-25 Vdc	0.5
7	-50 Vdc	3.0
8	104 Vac root mean square (rms) ± 3 percent, 400 Hz ± 3 percent	0.2 rms

* The regulated secondary power voltage load currents are part of the load capacity specified in 3.2.1.11.1

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TABLE III

LFTS ELECTRICAL POWER

Item	Input Power Form	Maximum Load
1	+29 ± 4 Vdc	0.6 kilowatt (kW)
2	115 Vac, 400 Hertz (Hz), 3-phase, 4-wire, Type III	9.0 kilovolt Amperes (kVA)
3	115 Vac, 60 Hz, 1-phase, Type I	1.75 kVA
4	115 Vac, 60 Hz, 3-phase, 3-wire, ungrounded, Type I	4.4 kVA

3.2.2.2 Clean. air. Applicable. The LFTS requirements for clean air shall not exceed 45 pounds per minute.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Applicable.

3.2.2.5 Helium. Applicable.* Provisions shall be made for connection to a utility line.

3.2.2.6 Nitrogen. Applicable.

3.2.2.7 Hydraulic fluid. Applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. Applicable. Electrical power required for operation of the LFTS shall be as listed in Table III.

3.2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Not applicable

3.2.3.2.2 intermediate level. Applicable.

3.2.4 Maintainability. Applicable except as follows.

3.2.4.1 Mean time to repair, Applicable.

* Used with OJ-215( )/AWM-23(V) only.

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3.2,4.2 Test points. Applicable. Test points shall be provided to facilitat fault Isolation to the SRA level, fault isolation within an SRA, alignments or adjustments, calibration, and signal monitoring. Location and accessibility of test points shall be-dictated-by consideration of ease of maintenance and repair,

3.2.4.3 Shop replaceable assemblies (SRA). Applicable.

3.2.4.4 Self test capabilities The LFTS shall provide capability to establish its functional readiness, Time required for self test shall not exceed 30 minutes. In the event of a self test failure indication, the LFTS shall provide the capability to isolate faults to:

a. The instrument level for commercial test equipment.
b. The SRA level for peculiar equipment.

3.2.5 Reliability. The mean time between failures shall be 125 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagentic Interference (EMI). Applicable, with the following exceptions:

a. MIL-STD-461, tests CE03 and RS02-1 and the limits therein shall apply.
b. MIL-STD-461, tests CE02 and CE04 shall apply whenever power remote control leads external to the station are used.
c. MIL-STD-461, tests CE01, RS01, RS02-2, and RS04 shall not apply.

3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection Applicable, except that test station ‘protection shall be provided for an out of tolerance flow rate only.

3.2.7.2 Clean air protection. Applicable.

3.2.7.3 Electrical power protection, Applicable.

3.2.7.3,1 Secondary voltage protection. Occurrence of an out-of-tolerance secondary voltage condition that could damage either an item under test or the test station shall result in automatic shutdown of the secondary power forms. The out-of-tolerance condition leading to shutdown shall be identified by an appropriate indicator. Operator action shall be required to reinstate test station operation after the occurrence of automatic shutdown.

3.2,7. 3.2 Emergency shutdown switches, Emergency shutdown switches shall be provided for use by the operator. When an emergency shutdown switch is

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activated, the load side of all LFTS primary circuit breakers shall be disconnected from the respective loads. The switches utilized for this application shall be of the pushbutton-activated, momentary type with the pushbutton finished in red. The front panel area around the pushbutton shall be marked with alternating orange-yellow and black stripes. A switch shall be centrally located on the LFTS in an area which is easily accessible to the operator during normal operation and another switch shall be located on the antenna mounting stand.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching and coolant control. The LFTS shall provide the necessary switching to control power forms of 3.2.2 and baffling as required for control of coolants of 3.2.2.

3.2.10 Displays. Visual displays shall provide an indication of the test status of the unit under test and the LFTS.

3.3 Performance.

3.3.1 General. The LFTS shall provide the capability to test and maintain the equipment listed in Table IV.

3,3.2 Organizational level functional requirements. Not applicable.

TABLE IV

WRAs MAINTAINED AT THE LFTS

Item	WRA Nomenclature (See 6.4)
1	Antenna AS-2764( )/AWG-9
2	Servo Electronic Subassembly (Part of AS-2764( )/AWG-9)
3	Control, Antenna - Test C-9038( )/AWG-9 and C-10036( )/AWG-9
4	Receiver, Infrared R-1786( )/AWG-9*
5	Processor, Data CV-2947( )/AWG-9*
6	Power Supply PP-6773( )/AWG-9
7	Power Supply PP-6774( )/AWG-9
8	Power Supply PP-7010( )/AWG-9 and PP-7042( )/AWG-9
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3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warmup time. Applicable.

3,4 Detailed functional requirements.

3,4.1 Test capabilities. The LFTS shall provide and control the stimuli, the power forms, and the loads required to perform functional and diagnostic testing of the units listed in Table IV. The test capabilities provided by the LFTS shall be adequate to verify proper operation of the item under test.

3.4.1.1 Antenna AS-2764( )/AWG-9. The LFTS shall be capable of performing the following tests on the radar antenna:

a. Power and interlock test.

(1) Verify power
(2) Verify hydraulic interlock

b. Pointing tests.

(1) The antenna azimuth and elevation positions shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 40 degrees.
(2) The IR slave capability shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 40 degrees.

c. Display resolver tests.

(1) The antenna azimuth and elevation search mode positions shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 40 degrees.
(2) The antenna azimuth and elevation track modes $_1{}_e$ and j_e shall be measured to an accuracy of ± 1 Percent in the range from O to 10 Vat.

d. Gimbal limit tests. The azimuth and elevation gimbal limit angles shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 65 degrees.
e. Azimuth and elevation search response tests. The azimuth and elevation search response band pass shall be measured to an accuracy of ± 4 percent $(\theta_{out}/\theta_{1n})$ at a frequency band to 9.5 Hz.
f, Azimuth and elevation track response tests. The azimuth and elevation track response band pass shall be measured to an accuracy of ± 4 percent $(\theta_{out}/\theta_{in})$ in the range from 1 to 1.6 Hz.
g. Ferrite modulator tests. The azimuth and elevation modulation current shall be 138 milliamperes ± 2 percent.
h. Boresight ranging tests.

(1) The square wave amplitude shall be measured to an accuracy of ± 3 percent in the region of 20 V p-p.

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i. Built-in-test (BIT) tests.

(1) The BIT pointing test capability shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 30 degrees.
(20 The azimuth and elevation BIT search fault isolation test (FIT) signal shall be measured to an accuracy of ± 1 percent in the range from 0 to 5 Vdc.
(5) The azimuth and elevation BIT track fault isolation test signal shall be measured to an accuracy of ± 1 percent in the range from 0 to S Vdc.

j. Antenna pre-position tests, The antenna azimuth and elevation position shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of -40 degrees elevation and O degree azimuth

1.2 Servo electronic subassembly. The LFTS shall be capable of perg the following tests on the servo electronic subassembly of Antenna 4( )/AWG-9.

a. Power regulator tests.

(1) The dc voltages shall be measured to an accuracy of ± 0.5 percent in the range from ± 12 to ± SO Vdc.
(2) The dc voltage ripple shall be measured to an accuracy of ± 3 percent in the region of 150mV.
(3) The power logic shall be verified.

b. Amplifier/phase detector gain test. The azimuth and elevation amplifier gain shall be measured to an accuracy of ± 4 percent for both search and track.
c. Servo response test. The azimuth and elevation valve driver response shall be measured to an accuracy of ± 10 percent.
d. Tachometer loop response tests, The azimuth and elevation amplifier response gain and band pass shall be measured to an accuracy of ± 4 percent (θ_{out}/θ_{in}).
e. Secant multiplier tests. The azimuth and elevation error signal shall be measured to an accuracy of ± 1 percent for three levels of search and track in the range from O to 6.5 Vdc ,
f. Gimbal limit tests, The azimuth and elevation gimbal limit angles shall be measured to an accuracy of ± 3 percent ± 0.2 degree in the region of ± 65 degrees.
g. Display feedback resolver excitation tests The azimuth and elevation resolver excitation (η and ε) shall be measured to an accuracy of ± 5 percent in the range from 0 to 10 Vrms.
h. Spin motor drive tests.

(1) The azimuth drive sum amplitude shall be measured to an accuracy of ± 2 percent in the region of 6.5 Vat.

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(2) The elevation drive sum amplitude shall be measured to an accuracy of ± 2 percent in the region of 6.5 Vat.

l. Position signal switching.

(1) The antenna servo command switching shall be verified.
(2) The antenna pointing switching shall be verified.
(3) The antenna pre-position switching shall be verified
(4) The IR slave switching shall be verified,

j. Built-in test (BIT). The azimuth and elevation FIT signal shall be measured in both search and track to an accuracy of ± 1 percent.

k. Solenoid and hydraulic interlock. switching shall be verified.

3.4.1.3 Control, Antenna - Test C-9038( )/AWG-9 and C-10036( )/AWG-9. The LFTS shall be capable of performing the following tests on the antenna and the antenna test control.

a. Pointing test - computer controlled mode.

(1) The azimuth and elevation angles shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 40 degrees.
(2) η_S and ε_S shall be measured to an accuracy of ± SO mVdc in the range from O to ± 10 Vdc.

b. Pointing test - panel controlled mode.

(1) The azimuth and elevation angles shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of ± 40 degrees.
(2) η_S and ε_S shall be measured to an accuracy of ± 50 mVdc in the region of 150 mVdc.

c. Pointing test - hand controlled mode. The azimuth and elevation angles shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of -40 degrees.

d. Scan rate and frame time.

(1) η' shall be measured to an accuracy of ± 1 percent in the region of 1 Vdc.
(2) η shall be measured to an accuracy of ± 1 percent in the range from 0.5 to 1.6 seconds.

e. Antenna bar step tests.

(1) ε_S shall be measured to an accuracy of ± 0.5 percent in the computer controlled 4 bar scan,
(2) The radar elevation bar analog shall be measured to an accuracy of ± 0.5 percent in the manual 8 bar scan.

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f. Modulator tests.

(1) In modulator tests F1, F2, and F3 the modulation current shall be measured to an accuracy of ± 5 percent.
(2) In modulator tests F1, F2, and F3 the phase reference shall be measured to an accuracy of ± 10 usec in the range from 500 to 2500 usec

g. Modulator tests, random. The azimuth and elevation modulation current shall be measured to an accuracy of ± 3 percent.

h. Azimuth and elevation static gain and crosstalk tests.

(1) The gain shall be measured to an accuracy of ± 3 percent in the region of 6.5 Vdc.
(2) The crosstalk shall be measured to an accuracy of ± 3 percent in the region of 100 mVdc.

i. Track response tests. The azimuth and elevation bandwidth shall be measured to an accuracy of ± 2 percent in the range from 0.1 Hz to 10 Hz.

j. Gyro cage tests, The azimuth and elevation gain shall be measured to an accuracy of ± 2 percent,

k. Feedback and feedforward tests.

(1) The azimuth and elevation gain shall be measured to an accuracy of ± 2 percent.
(2) The azimuth and elevation feedforward square wave amplitude shall be measured to an accuracy of ± 3 percent.

1. Position memory tests. The ω_d and ω_e drift shall be measured to an accuracy of ± 1 percent in the range from O to 10 Vdc.

m. Angle rate memory tests. The ω_d and ω_e drift shall be measured to an accuracy of ± 1 percent in the range from O to 10 Vdc.

n. Rate memory tests.

(1) ω_d and ω_e shall be measured to an accuracy of ± 1 percent in the range from O to 10 Vdc.
(2) The ω_d drift and the ω_e drift shall be measured to an accuracy of ± 1 percent in the region of 10 Vdc.

o. Display position error tests. For azimuth and elevation slew rate, the rate output shall be measured to an accuracy of ±2 percent in the region of 29 V p-p,

p. Azimuth memory tests. The azimuth drift shall be measured to an accuracy of ± 0.1 degree/second.

q. BIT tests. For azimuth and elevation, the ω_d and ω_e shall be measured to an accuracy of ± 0.5 percent in the range from O to 10 Vdc.

r. BIT counter input tests.

(1) The type A digital signal shall be verified.

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(2) The signal pulse width shall be measured to an accuracy of ± 0.1 usec.

s. BIT modulation tests.

(1) The frequency shall be measured to an accuracy of ± 1 Hz.
(2) The noise modulation shall be measured to an accuracy of ± 3 percent.
(3) The amplitude shall be measured to an accuracy of ± 3 percent.
(4) The LPRF signal pulse width, rise and fall time, and pulse jitter shall be measured to an accuracy of ± 0.01 usec in the region of 0.05 usec.

t. BIT target generation tests.

(1) The target frequency shall be measured to an accuracy of ± 1 Hz.
(2) The target amplitude shall be measured to an accuracy of ± 3 percent,

u. BIT 3.2 megaHertz (MHz) target generation tests.

(1) The target frequency shall be measured to an accuracy of ± 1 Hz.
(2) The target amplitude shall be measured to an accuracy of ± 3 percent.

v. BIT horn target insertion control tests.

(1) The target amplitude shall be measured to an accuracy of ± 3 percent.
(2) The rise and fall time shall be measured to an accuracy of ± 0.01 usec.

w. BIT reference oscillator tests,

(1) The frequency shall be measured to an accuracy of ± 1 Hz.
(2) The amplitude shall be measured to an accuracy of ± 3 percent.

x. Logic and analog tests.

(1) The line driver functions shall be verified.
(2) The line receiver functions shall be verified.
(3) The standard serial interface (SSI) shall be verified.
(4) The analog switching shall be verified.

3.4.1.4 Receiver, Infrared R-1786( )/AWG-9.* The LFTS shall be capable of performing the following tests on the infrared (IR) receiver:

* Tested on OJ-215( )/AWM-23(V) only.

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a. Pointing tests. The pointing angles shall be measured to an accuracy of ± 1 percent ± 0.2 degree in the region of 0 and ± 40 degrees.
b. Response tests. The azimuth and elevation motor drive response gain and band width shall be measured to an accuracy of ± 2 percent in the 1 to 10 Hz range.
c. Detection tests.

(1) The detector cool down time shall be verified by measuring the detector noise to be less than 60 millivolts peak-to-peak in less than 5 minutes,
(2) The clock pulse signal shall be measured in PRF and amplitude to an accuracy of ± 3 percent at 200 kiloHertz (kHz) in the region of 2.5V peak.
(3) The staircase signal shall be measured to an accuracy of ± 3 percent in the region of 1.4V at 25kHz.
(4) The multiplexed video shall be measured to an accuracy of ± 10 percent in the region of 150 mV peak.

d. BIT source test. The multiplexed video target shall be measured to an accuracy of ± 3 percent in the region of 100 mV.

3.4.1.5 Processor, Data CV-2947( )/AWG-9.* The LFTS shall be capable of performing the following tests on the IR data processor.

a. Pointing tests.

(1) The position output shall be measured to an accuracy of ± 1 .0 percent ± 0.2 degree in the region of ± 40 degrees.
(2) η_s expanded sweep IR shall be measured to an accuracy of ± 0.1 percent in the region of 11.5 Vdc.
(3) ε_s IR shall be measured to an accuracy of ± 0.1 percent in the region of 11.5 Vdc.

b. Azimuth scan tests.

(1) $\dot{\eta}_s$ and frame time shall be measured to an accuracy of ± 1 percent for the ± 65, ± 40, ± 20 and ± 6 degree scans in the region of 8 Vdc.
(2) The cone angle shall be measured to an accuracy of ± 0.7 percent ± 0.2 degree in the range from +55 to +61 degrees.

c. Hand control and elevation bar step tests. ε_s IR shall be measured to an accuracy of ± 0.1 percent in the hand controlled 8, 4, and 2 bar scan in the range from 0 to 12 Vdc.
d. Display video and aural tone tests.

(1) The aural tone in search mode and full action mode shall be verified,
(2) The IR staircase plus video shall be measured to an accuracy of ± 3 percent in the region of 2.8 V peak.

* Tested on OJ-215( )/ANM-23(V) only.

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(3) The IR display video shall be measured to an accuracy of ± 3 percent in the region of 5.4 V peak.
(4) The IR azimuth tracking gate shall be measured to an accuracy of ± 3 percent in the region of 7.5 V peak.

e. Track circuits tests.

(1) η_s expanded sweep IR shall be measured to an accuracy of ± 2 percent in the region of 1 Vdc.
(2) The azimuth track position signal shall be measured to an accuracy of ± 2 percent in the region of 1 Vdc.
(3) The detector number valid signal shall be measured to an accuracy of ± 3 percent in the region of 7 V peak.

f. Azimuth tracking 0 degree, elevation tracking ± 25 degrees tests.

(1) IR track valid shall be verified.
(2) IROT shall be verified,
(3) IR track shall be verified,
(4) ε_s IR shall be measured to an accuracy of ± 2 percent in the region of 1 Vdc.
(5) Azimuth track position shall be measured to an accuracy of ± 100 mVdc.
(6) Return to search shall be verified.

g. Raid size estimate and IR not cool tests,

(1) IR computer video No. 1, No. 2, and No. 3 shall be verified,
(2) IR not cool signal shall be verified.
(3) IR cool signal shall be verified.

h. Servo tests.

(1) The elevation motor drive shall be measured to an accuracy of ± 100 mVdc.
(2) The azimuth motor drive shall be measured to an accuracy of ± 100 mVdc.
(3) Cos ε IR shall be measured to an accuracy of ± 0.1 percent in the region of 10 Vdc.
(4) The servo gain shall be measured to an accuracy of ± 2 percent in the gain region of 100 to 200,

i. Slave tests,

(1) Video blanking shall be measured to an accuracy of ± 3 percent in IR slave, radar scan mode and radar track mode in the region of 8.5 V peak.
(2) ε_s IR shall be measured to an accuracy of ± 100 mVdc in IR slave, radar scan mode,
(3) IR slaved shall be verified in IR slave, radar scan mode.

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(4) Azimuth and elevation angles shall be measured to an accuracy of ± 1.0 percent ± 0,2 degree in IR slave, radar scan mode in the region of +40 degrees.
(5) n_s expanded sweep IR signal slope shall be measured to an accuracy of ± 1 percent in region of 1 Vdc and period to an accuracy of ± 1 percent in the region of 0.3 second in IR slave radar track mode.
(6) The azimuth motor drive shall be measured to an accuracy of ± 3 percent in IR slave, radar track mode in the region of 25 Vrms.
(7) In the radar slave, IR track sample mode the azimuth and elevation angles shall be measured to an accuracy of ± 1.0 percent ± 0.2 degree in the region of 35 to 45 degrees.
(8) In the radar slave, IR track sample mode the i_{dc}, j_{dc}, and k_{dc} outputs shall be measured to an accuracy of ± 0.1 percent in the region of 5 to 6 Vdc.
(9) In the radar slave, IR track hold mode the azimuth and elevation angle shall be measured to an accuracy of ± 1.0 percent ± 0.2 degree in the region of 35 to 45 degrees.
(lo) Return to search shall be verified.

j. BIT tests.

(1) The IR display video shall be measured to an accuracy of ± 3 percent in raid size estimate test in the region of 5.4 V peak,
(2) The azimuth and elevation angles shall be measured to an accuracy of ± 1.0 percent ± 0.2 degree in the BIT search test in the region of O degree.
(3) The elevation scan shall be measured to an accuracy of ± 5 percent in the region of ± 30 degrees.
(4) The BIT search error gain shall be measured to an accuracy of ± 2 percent in the region of 3.4 V peak.
(5) The azimuth and elevation motor drive shall be measured to an accuracy of ± 100 mVdc.
(6) The BIT source voltage return shall be verified.

k. Power output tests.

(1) The dc voltages shall be measured to an accuracy of ± 0.1 percent in the 10 to 50 Vdc range.
(2) The ac voltages shall be measured to an accuracy of ± 1 percent in the 100 to 120 Vac range and in the region of 28 Vat.
(3) The returns shall be verified.
(4) IR on board shall be verified.

3.4.1.6 Power Supply PP-6773( )/AWG-9. The LFTS shall be capable of performing the following tests on the semiregulated dc power supply:

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a. The dc voltages shall be measured to an accuracy of ± 1 percent in the 0 to 360 Vdc range.
b. The dc voltage ripple shall be measured to an accuracy of ± 3 percent in the range of 0 to 10 V p-p.

3.4.1.7 Power Supply PP-6774( )/AWG-9, The LFTS shall be capable of performing the following tests on the regulated 28 Vdc, 104 Vac power supply,

a. IR regulated voltage tests,

(1) The regulated voltages shall be measured to an accuracy of ± 0.5 percent in the 5 to 50 Vdc range.
(2) The dc voltage ripple shall be measured to an accuracy of ± 3 percent in the 150 mVdc region.

b. Missile auxiliary regulated voltage tests.

(1) The regulated voltages shall be measured to an accuracy of ± 0,25 percent in the 5 to 110 Vdc range.
(2) The regulated voltage ripple shall be measured to an accuracy of ± 3 percent in the 300 mVdc region.

c. 28 Vdc unregulated voltage tests,

(1) The 28 Vdc shall be measured to an accuracy of ± 3 percent.
(2) The 28 Vdc ripple shall be measured to an accuracy of ± 3 percent.

d. 104 Vat, 400 Hz reference voltage tests.

(1) The 104 Vac shall be measured to an accuracy of ± 1 percent.

e. Logic and timing shall be verified.
f. BIT fault signals shall be verified.
g. Regulator current limiting shall be verified.

3.4.1.8 Power Supply PP-7010( )/AWG-9 and PP-7042( )/ANG-9. The LFTS shall be capable of performing the following tests on the missile power supply:

2. Launch missile-on-aircraft-test (MOAT) power station 2 through station 7 tests.

(1) Voltage full load. The voltage of launch. MOAT power A and launch MOAT power B at full load shall be measured to an accuracy of ± 1 percent in the region of +97 Vdc.
(2) Overload protection shall be verified.
(3) BIT fault shall be verified.
(4) Diode short test.
(5) Current full load. The current of launch MOAT power A and launch MOAT power B at full load shall be measured to an accuracy of ± s percent in the region of 4.0A.

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b. Ready launch MOAT (RLM) power, station 2 through station 7 tests.

(1) Voltage full load. The high load voltage shall be measured to an accuracy of ± 1 percent in the +97 Vdc region.
(2) Current full load. The high load current shall be measured to an accuracy of ± 5 percent in the d region.
(3) Overload protection shall be verified.
(4) BIT fault shall be verified.
(5) Power reset shall be verified
(6) Diode short test.

c. Sidewinder expanded acquisition mode (SEAM).

(1) SEAM logic shall be verified.
(2) SEAM processing shall be verified.

3.5 Workmanship. Applicable.

3.6 Elements.

3.6.1 Console, Test, Low Frequency, Receiver-Ampllfler OJ-253( )/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.1.1 Panel, Power Distribution SB-3696( )/AWM-23(V), The power distribution panel shall provide the capability to control and monitor the utility and secondary power forms in the test station. In addition, protection and safety control shall be provided for personnel and equipment in this panel. Voltage form monitoring (utilities and secondary), liquied coolant monitoring, cooling air monitoring, and malfunction indication capabilities shall be provided by this panel.

3.6.1.2 Counter/timer. A counter/timer, Hewlett-Packard S326B/001/011 or equivalent, shall be supplied and shall provide the capability to perform time interval and frequency measurements.

3.6.1.3 Digital multimeter. A digital multimeter, Systron-Dormer 9340-5B, or equivalent, shall be supplied and shall, provide the capability to perform ac and dc voltage, and resistance measurements.

3.6.1.4 Monitor, Coordinate Data ID-1861( )/AWM-23(V).* The Coordinate data monitor panel shall provide the following major items for testing the antenna, IR receiver, IR amplifier, and the antenna controller:

a. An Instrument servo to provide a visual indication of pointing angles for the antenna servo and IR WRAs.
b. An audio speaker for monitoring the aural tone from the IR.
c. A reference dc power supply to excite the azimuth and elevation controls for generation of η and ε.

* IR WRAs are tested on OJ-215( )/AWM-23(V) only.

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3. (),1.5 Frequency response analyzer. A frequency response analyzer, EMR Hatboro 1410AR-1, or equivalent, shall be supplied to provide the capability to perform loop response measurements,

3.6.1.6 Panel, Coolant Supply MX-9307( )/AWm-23(V).* The coolant supply panel shall provide electrical connections for the anterna servo electronic subassembly, cooling air connections for Control, Antenna-Test C-9038( )/AWG-9, electrical and liquied coolant connections for Receiver, Infrared R-1780( )/AWG-9, and liquid coolant connection for Processor, Data CV-2947( )/AWG-9.

3.6,1.7 Mounting MT-4502( )/AWM-23(V).* The mounting shall provide electrical connections for the Control, Antenna-Test C-9038( )/AWG-9 and Processor, Data CV-2947( )/AWG-9. The mounting shall also provide mounting for the Control, Antenna-Test, the Processor, Data, the IR receiver/cryogenic refrigerator fixture, and the antenna servo fixture.

.3,(). 1.8 Power Supply PP-6850( )/AWM-23(V), The power supply shall provide the capability to supply dc voltages to WRAs and LFTS circuits, using another dc power supply (5,6.1.9) as a source. The power supply shall also provide the capability to sense the output power forms for overvoltage, undervoltage, and overcurrent for LFTS and WRA protection as well as for continuous monitor capability

3.6.1.9 Power supply, dc. A dc power supply, Lambda LB706-FM-3632, or equivalent , shall be supplied and shall provide 200 Vdc at 10 amperes to Power Supply PP-6850( )/AWM-23(V) [3.6.1.8).

3.6. 1.10 Distribution Box J-3110( )/AWM-23(V). The distribution box shall control and distribute power to units in the test station.

3.6.2 Console, Test, Low Frequency, Antenna OJ-254( )/AWM-23(V) and OJ-380( )/AWM-23(V)**. The equipment of this console shall be as specified in the following

3.6.2.1 Oscilloscope. An oscilloscope, Hewlett-Packard 181AR with 1801A-003 and 1821A plug-ins, or equivalent, shall be supplied to provide display and observatino of various wave forms.

3.6.2.2 Panel, Test, Electrical C-9179( )/AWM-23(V). The electrical test panel shall provide the controls, displays, and switching capabilities to conduct functional and fault isolation tests on the WRAS assigned to the LFTS. Controls on the front panel shall be provided to command WRA Power On or Power Off, command emergency for off station power, select and enable functional or fault isolation tests, select and apply input signals to WRAs, select measurement instruments during tests, command SSI word generator to provide desired word, control setting of potentiometers, and control level of simulated WRA inputs. Displays on the front panel shall be provided to display test number, test mode and status, and SSI BIT fault.

* IR WRAs are tested on OJ-215( )/AWM-23(V) only.
** OJ- 254 is part of OJ-215( )/AWM-23(V) only.
OJ-380 is part of OJ-379( )/AWM-23(V) only.

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3,6.2.3 Switching Unit, Fault Isolation SA-1895( )/AWM-23(V). The fault isolation switching unit shall provide the radio frequency and low frequency matrices required to switch the test signals to the monitoring instruments during fault Isolation. WRA fault isolation selector assembly shall also be included in this panel. The test controller and test circuits drawer shall provide the switching, logic, and RF circuits necessary for testing the test controller section of the Control, Antenna-Test C-9038( )/AWG-9 WRA. The 32 bit SSI words from the WRA storage registers shall be switched to WRA line receiver inputs and to drawer logic in-order to test the storage register and analog WRA circuits. Logic circuits shall be included in the drawer to provide GO/NO-GO signals to read out indicators on the electrical test panel (3.6.2.2). An RF amplifier shall be provided in the drawer to amplify low level signals from the RF section of the WRA for frequency counting. RF switching circuits shall also be included for switching signals to monitoring Instruments. The panel shall also provide the capability to null the antenna servo components.

3,6.2.4 Switching Unit, Fault Isolation SA-1896( )/AWM-23(V) and SA-2086( )/AWM-23(V)*. The fault isolation switching unit shall provide the stimulus circuits and the load circuits for the antenna servo and IR WRAs. This drawer - shall also provide the switching matrices for WRA signal switching to the WRA and from the WRA for functional test and fault isolation.

3.6.3 Console, Test, Low Frequency, Power Supply OJ-255( )/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.3.1 Spectrum analyzer. A spectrum analyzer, Hewlett-Packard 141T with 8553B and 8552A plug-ins, or equivalent, shall be supplied. The spectrum analyzer shall provide the capability to measure the RF spectrum of the test controller portion of the Control, Antenna-Test C-9038( )/AWG-9.

3.6.3.2 Switching Unit, Fault Isolation SA-1897( )/AWM-23(V). The fault isolation switching unit shall provide the stimuli circuits required for testing Power Supply WRAs PP-6773( )/AWG-9, PP-6774( )/AWG-9, PP-7010( )/AWG-9 and PP-7042( )/AWG-9. This panel shall also provide the switching matrices for switching the power supply outputs to the monitoring instruments for functional test and fault isolation of the power supply WRAs.

3.6.3.3 panel, Coolant Supply MS-9308( )/AWM-23(V). The coolant supply panel shall provide cooling air connections for Power Supply WRAs PP-6773( )/AWG-9, PP-6774( )/AWG-9, PP-7010( )/AWG-9 and PP-7042( )/AWG-9.

3.6.3.4 Mounting MT-4503( )/AWM-23(V). The mounting shall provide mounting for Power Supply WRAs PP-6773( )/AWG-9, PP-6774( )/AWG-9, PP-7010( )/AWG-9 and PP-7042( )/AWG-9. The mounting surface shall also provide electrical connections for WRAs under test.

* SA-1896 is part of OJ-215( )/AWM-23(V) only.
SA-2086 is part of OJ-379( )/AWM-23(V) only.
IR WRAs are tested on OJ-215( )/AWM-23(V) only.

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3.6.3.5 Dummy Load, Electrical DA-631( )/AWM-23(V) and DA-632( )/AWM-23(V). The dummy loads shall provide loading for Power Supply WRAs PP-6773( )/AWG-9, PP-6774( )/AWG-9, PP-7010( )/AWG-9 and PP-7042( )/AWG-9 during functional test and fault isolation. The dummy loads shall provide high-dissipation load resistors cooled by liquid coolant.

3.6.4 Stand, Mounting, Antenna MT-4445( )/AWM-23(V). The antenna mounting stand shall serve as a holding fixture for the Antenna AS-2764( )/AWG-9 WRA when mounted in Adapter, Handling, Antenna MX-9046( )/AWM-23(V).

3.6.5 Accessories - LFTS, The LFTS accessories shall include the following items which are not permanently attached to the LFTS.

3,6.5.1 Extenders - LFTS. Extender cards and cables shall provide the capability to extend modules from their respective WRAs or GSE assemblies.

3.6-.5.2 Fixture, pressure leakage - test. The pressure leakage test fixture shall provide capability and pressure to test waveguides for leakage,

3.6.5.3 Fixture, receiver/refrigerator cryogenic-infrared. * The infrared receiver/refrigerator cryogenic fixture shall serve as a holding fixture for Receiver, Infrared R-1786( )/AWG-9.

3.6.5.4 Fixture, antenna servo. The antenna servo fixture shall serve as a holding fixture for the servo electronic subassembly of Antenna AS-2764( )/ AWG-9.

3.6.5.5 Fixture, refrigerator test.* The refrigerator test fixture shall provide test capability for the refrigerator subassembly pf the Receiver, Infrared, R-1786( )/AWG-9.

3.6.5.6 Equipment, helium filling.* The helium filling equipment shall provide the necessary controls to replenish the helium of 3.2.2.5 in the refrigerator subassembly of Receiver, infrared R-1786( )/AWG-9. The helium filling equipment shall include a fitting for recharging.

3.6.6 Elapsed time meter. An elapsed time meter conforming to MIL-M-7793 shall be provided. The meter shall be readily visible during operation of the LFTS and shall be capable of registering at least 9,999 hours.

4. QUALITY ASSURANCE PROVISIONS

4.i Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classifications. Applicable

4.2.2 individual items. Applicble

* Used with OJ-215( )/AWM-23(V) only.

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.2.3 In-process Inspection. Applicable

.2.4 In-process data. Applicable.

.2.5 Acceptance test plan. Applicable. Table V shall apply.

.2.6 Test results. Applicable.

.2,7 Acceptance. Applicable.

.2.8 Utilities. Applicable.

.2.9 Configuration. Applicable.

.3 Tests.

.3.1 Electrical and electronic items - operational run-in test. Applicable.

.3.2 Mechanical items. Applicable.

.4 Environmental test conditions. Applicable.

.5 Test location, Applicable.

.6 Acceptance schedule. Applicable.

.7 Test procedures. Applicable.

.8 Rejection and retest. Applicable.

PREPARATION FOR DELIVERY

.1 General. Applicable.

NOTES

.1 Intended use. The LFTS is intended for intermediate level use to ctionally test, fault isolate, and verify the serviceability of certain pon replaceable assemblies (WRAs) of the infrared,* missile auxiliary, ar, and power supply subsystems of Weapon Control System (WCS) AN/AWG-9.

.2 Ordering data. Applicable,

.3 Definitions. Applicable.

.4 Nomenclature cross reference. The following is a cross reference ween the official Navy nomenclature and the major contractor part numbers. contractor part number suffix is controlled by the configuration list the contract.

IR WRAs are tested on OJ-215( )/AWM-23(V) only.

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TABLE V

ACCEPTANCE TEST PLAN FOR LOW FREQUENCY
TEST STATION OJ-215( )/AWM-23(V) AND
OJ-379( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate self test and operational run-in tests of the Low Frequency Test Station.

2.0 TEST CONDITIONS

a. AWG-9 Slave Units (none)
b. Utilities Required (Information only, 4.2.8)

(1) Power as specified in 3.2.2.1
(2) Clean air cooling as specified by 3.2.2.2
(3) Liquid cooling as specified by 3.2.2.4

3.0 TESTS in accordance with Acceptance Test Procedure 3282206

a. Perform self-tests
b. Perform operational run-in tests

	Official Nomenclature	Contractor Part No.
a.	Antenna AS-2764( )/AWG-9	481031-1XX
b.	Control, Antenna - Test C-9038( )/AWG-9 and C-10036( )/AWG-9	481081-1XX
c.	Receiver Infrared R-1786( )/AWG-9	481101-1XX*
d.	Processor, Data CV-2947( )/AWG-9	481120-1XX*
e.	Power Supply PP-6773( )/AWG-9	481601-1XX
f.	Power Supply PP-6774( )/AWG-9	481610-1XX
g.	Power Supply PP-7010( )/AWG-9 and PP-7042( )/AWG-9	481730-1XX
h.	Test Station, Low Frequency OJ-215( )/AWM-23(V) and OJ-379( )/AWM-23(L)	493350-1XX
i.	Console, Test, Low Frequency, Receiver-Amplifier OJ-253( )/AWM-23(V)	493351-1XX
j.	Console, Test, Low Frequency, Antenna OJ-254( )/AWM-23(V) and OJ-380( )/AWM-23(47)	493352-1XX
k.	Console, Test, Low Frequency, Power Supply OJ-255( )/AWM-23(V)	493353-1XX
l.	Stand, Mounting, Antenna MT-4445 ( )/AWM-23(V)	493354-1XX

* IR WRAs are tested on OJ-215( )AWM-23(V) only.

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m.	Target Simulator, Infrared SM-630 ((/AWM-23(V)	493505-1XX*
n.	Kit, Tool, Special TK-222 ( )/A WM-23(V)	493601-1XX
o.	Adapter, Handling, Antenna MX-9046( )/AWM-23(V)	493602-1XX
p.	Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V)	493711-1XX
q.	Service Set, Hydraulic System MX-9045( )/AWM-23(V)	493703-1XX
r.	Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V)	493715-1XX
s.	Test Set, Antenna, Boresight TS-3S68( )/AWM-23(V)	493720-1XX

6.5 Liquid cooler. Applicable.

6.6 Use of this specification This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs Of MIL-T-81778(AS) are the same as MIL-T-81734(AS) the word "applicable" is used in this specification. When the term "not applicable" is used in this specification it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met. See 6,6 of MI-T-81734(AS).

6.7 Abbreviations and acronyms. Applicable. In addition, the following shall apply:

FIT	Fault isolation test
IR	Infrared
LFTS	Low frequency test station
MHz	Megahertz
MOAT	Missile on aircraft test
RLM	Ready launch MOAT
SEAM	Sidewinder expanded acquisition mode
[illegible]	Radar Antenna Azimuth Angle
[illegible]s	Space - Resolved Radar Antenna Azimuth Angle
[illegible]	Radar Antenna Elevation Angle
[illegible]s	Space - Resolved Radar Antenna Elevation Angle
[illegible]in, [illegible]out	Phase Angle
[illegible]d	Azimuth Line of Sight Rotation Rate
[illegible]e	Elevation Line of Sight Rotation Rate
[illegible]e, [illegible]e	Phasing Voltage (400 Hz Display Resolver Outputs)
mV	Millivolts

6 S Convience outlet power. Applicable.

6 9 Alternate Configurations. The OJ-379( )/AWM-23(V) configuration is the same as the OJ-215( )/AWM-23(V) configuration except that the OJ-379 does not have the capability to test WRAs and SRAs of the Infrared (IR)Subsystem of the AN/AWG-9 WCS. The OJ-215 configuration can be converted to the OJ-379 configuration by the deletion of the 5 SRAs listed in TABLE VI below from the

[illegible] IR WRAs are tested on OJ-215( )/AWM-23(V) only.

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Fault Isolation Switching Unit SA-1896( )/AWM-23(V) thereby converting it to the SA-2086( )/AWM-23(V) and by omitting the following accessories of 1.2e

(3) Fixture, Receiver/Refrigerator Cryogenic-Infrared	3.6.5.3
(5) Fixture, Refrigerator Test	3.6.5.5
(6) Equipment, Helium Filling	3,6.5.6

TABLE VI Reference SRAs

Qty	Ref Des	Description	Part Number
1	8A5A6	EL Servo Electronics Module	3432718
1	8A5A7	AZ Servo Electronics Module	3213516-1
1	8A5A9	Target Gate Generator Module	3375260
1	8A5A11	Staircase Generator Module	3375209
1	8A5A12	RSE Count Valid Module	3375087

6.10 Changes from previous issue are not marked.

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-81778C(AS) Test Station, Low Frequency OJ-215()/AWM-23 and OJ-379()/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

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